

Is a Data Science Degree Worth It in the AI Era?

The already-in-it guide — the full scores by track, the six factors behind them, and what to do from where you already are.

8.2

AI EXPOSURE · ENTRY DATA ANALYST
where 10 is most at risk

For anyone already in it — in the degree, or job-hunting.



Before you read this

The choice is already made — you're in a data science degree or training, or out of one and into the job hunt. So this guide skips the "should you study data science" question and goes straight to the one that's actually live: given where you already are, what do you do about it?

It's written to you directly — whether you are the one in it, or a parent reading on their behalf. Where it says "you," a parent can simply read "them"; the moves are the same either way.

What this is. A facts document, not a pep talk. It sets out what's known about where data science sits in an AI economy, where the evidence is contested, and where we might be wrong — and then what to actually do from here.

The good news, up front. Already in it is not too late — not close. Data science isn't one destination, it's a wide spread of them, from 6.0 to 8.2, and almost none of your exposure is fixed yet. Where you land inside that range is decided mostly by moves still in front of you — chiefly, whether you steer toward the building-and-ownership end rather than the querying end. The second half of this guide is those moves.

And one thing specific to data science. This is the guide where our own score faces the strongest challenge — and it cuts in your favor. The Bureau of Labor Statistics projects this occupation growing 33.5% over the next decade — the fourth-fastest in the entire US economy — while our framework rates the entry tier among the most exposed work we measure. Both readings are defensible, and the tension between them is the most useful thing in this document to understand from where you stand: demand is real and rising, but it's demand for a moving target, and steering toward it is on you.

The short answer

This is not the safe technical career it was sold as three years ago — and it is also one of the fastest-growing occupations in the country.

Entry-level data analysis scores **8.2** out of 10 for AI exposure. ML and AI engineering scores **6.0**.

The people building AI systems are reasonably protected. The much larger number who query databases, build dashboards and produce analyses are not — and that is where almost all the

entry-level jobs are.

But demand is growing fast enough that the picture is genuinely contested, and the section on the strongest challenge to our score sets out why — which is why, from where you already are, the work now is climbing toward the protected end, not just landing anywhere.

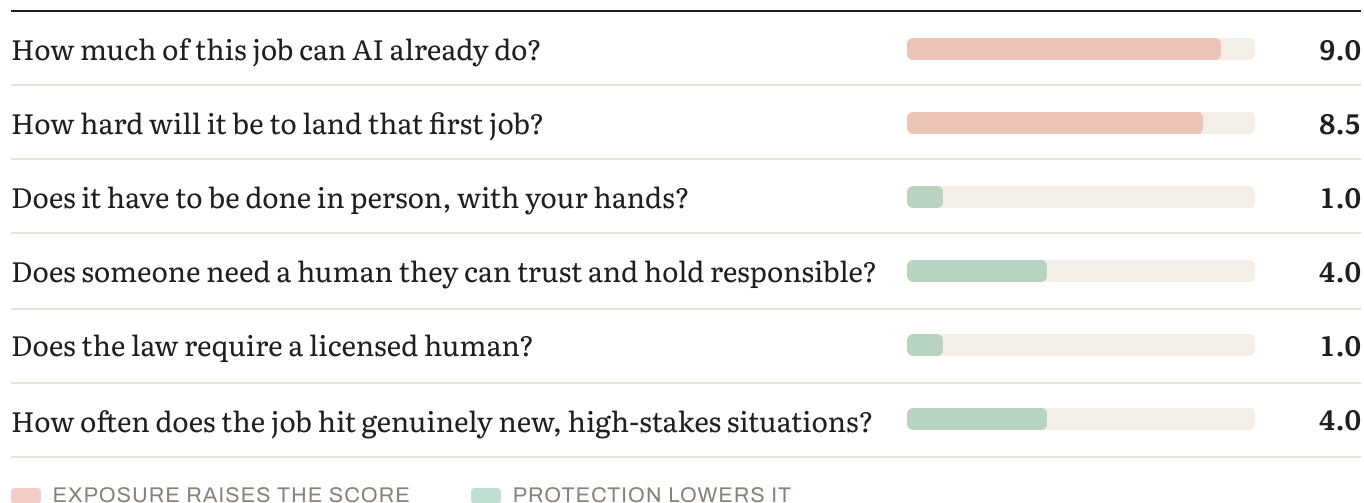
The scores

ROLE	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
ML / AI engineer	5.0	5.5	6.0	+1.0	Moderate
Senior data scientist	5.5	6.1	6.5	+1.0	Moderate-High
Data engineer	6.0	6.5	7.0	+1.0	Moderate-High
Analytics engineer / BI developer	6.4	7.1	7.6	+1.2	High
Entry data analyst	6.7	7.5	8.2	+1.5	High

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, entry-level software development scores 8.1, licensed engineering 4.0, and bedside nursing 2.8.

Read the table this way, and note what's missing. Nothing in this field scores below 6.0. There is no protected core here — only a protected ceiling. Every track moved at least a full point in three years, which no other profession in this index does. Your job, from here, is to move up the table toward the building end — not to hope the table moves.

Why data science scores high — the six factors



Here's how entry-level data analysis answers them. Knowing *why* the entry tier scores 8.2 while ML engineering holds 6.0 is what tells you which way to move.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

We rate this 9.0, and the reason is uncomfortable: data work is exactly the shape of task these systems handle best.

Writing SQL and queries. Building dashboards and reports. Exploratory analysis and summary statistics. Data cleaning and transformation. Standard model fitting and tuning. Documentation and analysis write-ups.

Inputs arrive as structured data. Outputs are code, charts and text. The section on proximity to AI explains why nearness to these systems offers no protection here.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Severe at entry, and this is where the evidence genuinely conflicts.

Industry analysis consistently reports the market favoring candidates with two to four years' experience — the same pattern documented in law, finance and consulting.

But entry-level salaries have risen sharply rather than fallen. Analyst starting pay is reported around \$90,000, up roughly \$20,000 in a year. That is not what a collapsing entry market looks like, and we take it seriously in the section on the strongest challenge to our score.

The way through isn't to wait for the generalist entry rung to soften. It's to enter *beside* it: aim at the engineering and ML end rather than analysis, build statistical depth that survives the automation of running the analysis, pair the work with a domain — healthcare, finance, energy — that lets you know when a result is wrong, and get real experience on messy data early,

because this is a field that prefers demonstrated experience over credentials more than almost any other. Those are concrete moves, not a hope, and the career-hunting checklist later in this guide lays them out.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

None of it. Zero, permanently.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

At senior level meaningfully; at entry level barely.

A senior data scientist telling an executive their strategy isn't supported by the data carries accountability that attaches to a person. An analyst producing a requested dashboard carries none — which means the fastest way to become protected is to get closer to a recommendation you answer for.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

No, and none is coming. No license, no reserved activity, no professional body controlling entry.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

At the ML engineering end frequently; at the analyst end rarely. Designing a system that will make decisions at scale has unprecedented failure modes. Producing a requested report does not. The distance between those two ends is the distance you're trying to close.

The strongest challenge to our own score

We normally put the counterargument at the end. Here it belongs before anything else, because it is the most serious one in this index and a reader deserves it up front.

The case against our 8.2:

- The Bureau of Labor Statistics projects data scientist employment growing from 245,900 in 2024 to 328,300 by 2034 — a 33.5% increase, and the fourth-fastest-growing occupation in the US economy. Roughly 23,400 openings a year.
- The World Economic Forum's 2026 Future of Jobs report ranked Big Data Specialists as the **single fastest-growing job in percentage terms** through 2030.
- McKinsey's State of AI found 88% of organisations now use AI in at least one business function, and named **data engineers and data scientists as the most in-demand AI**

hires, with supply lagging.

- BLS median pay sits around **\$112,590**, and reported entry-level analyst salaries have risen roughly \$20,000 in a year.

That last point deserves particular weight. If entry-level data work were being automated away, entry-level salaries would not be climbing. Prices generally fall when demand for something collapses.

Where we land, and why we hold the score anyway.

Our framework measures task exposure, not demand. Those are different things, and in data science they have come apart more than anywhere else we measure. A task can be highly automatable and highly in demand at the same time — which is precisely what a 33.5% growth rate alongside a 9.0 automatability rating describes.

What we think is actually happening: **the job title is surviving while its contents change completely.** A "data analyst" in 2030 will not do what one did in 2020, and the skill bar for the title has risen sharply — machine learning now appears in 77% of data scientist postings, and mentions in analyst postings have doubled.

The specific thing we're watching: whether entry-level postings requiring two years' experience or fewer stabilize or keep shrinking. If the genuine entry tier holds while demand grows, our 8.2 is too high and we will lower it. That is a falsifiable claim and we'll report it either way.

And there's a version of this guide that is simply optimistic, which a reader is entitled to hold: that data science is one of the best bets available, that the growth data is more reliable than our task analysis, and that we are pattern-matching from software to a field with different dynamics. We don't think that's right. We might be wrong.

Proximity to AI is not protection

There is an intuition worth dismantling here, because a lot of families are relying on it: that working with AI must be safer than working alongside it.

The index says otherwise, and the reason is structural rather than ironic.

Exposure is determined by the shape of the work, not by its subject matter. Our framework asks whether a task takes structured inputs, produces text or code as output, requires no physical presence, involves no licensed accountability, and repeats recognizable patterns.

Data analysis answers yes to every one — which is exactly what makes it a field where AI is useful, and exactly what makes it automatable.

A data analyst is not protected by understanding AI, any more than a copywriter is protected by understanding language.

What actually separates a 6.0 from an 8.2 inside this field is the same thing that separates tracks everywhere else: judgment and accountability.

An ML engineer designing a system that will make decisions at scale rates 8.0 on novelty, because the failure modes are unprecedented and expensive. An entry analyst producing a requested dashboard rates 4.0 on both novelty and accountability.

The general lesson: being close to a technology is not the same as being protected from it. Ask what shape the work is — where the inputs come from, whether a human must answer for the output — not what the work is about.

What the trend shows

Entry data analyst: 6.7 → 7.5 → 8.2. Up 1.5.

ML / AI engineer: 5.0 → 5.5 → 6.0. Up 1.0.

Every track moved at least a full point — which no other profession in this index does. In most fields the protected end barely shifts: bedside nursing moved 0.7, surgery 0.3. Here even the most protected role moved 1.0, because there is nothing structural holding any of it still.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Judgment under novelty	Real at ML engineering level	Eroding	Novel system design resists; standard modeling does not.
Human trust / accountability	Real at senior level	Durable	Someone must answer for a recommendation that costs money.
Regulatory / licensing	None	n/a	No license exists.
Physical / embodied	None	n/a	Permanently.

The honest read. Data science has two protections, both seniority-dependent, neither inherited from the field. Demand is doing far more work than structure here — and demand can change, while a license cannot. Which is why the direction you steer matters more than where you happen to start.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Writing the query	Deciding what question is worth asking
Building the dashboard	Interrogating a dashboard that arrived built
Cleaning the data	Verifying automated cleaning
Fitting the standard model	Judging whether the model is appropriate
Learning the business by assembling its numbers	Learning it... some other way
Recommending a decision and owning it	Unchanged

The pattern is now familiar across this index, and the fifth row is the recurring problem: analysts learned what a business actually did by building its numbers. That was tedious and it was the education. Nobody has replaced it — which means the people who go out of their way to learn what a business actually does, not just produce its numbers, have an edge that's about to get scarcer.

Human strengths that will matter most

1. **Deciding what question is worth asking** — the judgment upstream of any analysis, and the thing that doesn't automate. 2. **Knowing the data is wrong before the model does** — domain knowledge applied to numbers, and the most common cause of expensive mistakes. 3. **Owning a recommendation** — being the person who answers when a decision based on the analysis costs money. 4. **Understanding why the business cares** — analysts who understand the commercial question get to direct; those who don't get requests. 5. **Knowing what shouldn't be automated** — an increasingly valuable judgment.

Notably absent: SQL fluency, dashboard-building speed and breadth of tooling knowledge. These were the traditional markers of a strong junior analyst and are the fastest-depreciating. If your current value is mostly on that second list, the work is to shift it onto the first.

And underneath all five, two traits the score can't grade: **curiosity and adaptability**. The tooling here turns over faster than almost anywhere — the language, the library, the platform of the moment will all be replaced many times across a career. The people who compound are the ones who stay genuinely curious about the question rather than the technique, and re-tool without resenting it, rather than defending the stack they trained on. In a field with no license and no stable toolkit, that flexibility is itself a form of protection — and increasingly it's what employers hire for: the trajectory toward judgment and building, not the current tooling fluency.

Other things to consider about this job

What's genuinely good about it

The demand data is genuinely excellent, and it deserves stating clearly rather than grudgingly.

- **33.5% projected growth** through 2034 — fourth-fastest occupation in the US economy
- **~23,400 openings annually**, against a base of around 245,900 jobs
- **BLS median \$112,590**, with Glassdoor reporting averages substantially higher for experienced roles
- **Entry-level analyst salaries reported around \$90,000 and rising**
- Named by McKinsey among the most in-demand AI hires, with supply lagging

- Ranked by the World Economic Forum as the fastest-growing job in percentage terms through 2030

What people in the field report as rewarding: genuinely solving problems rather than producing outputs; unusual breadth of exposure to how a business works; fast feedback; and work that sits at the intersection of technical and commercial rather than purely one or the other.

And the specialization premium is steep. Machine learning appears in 77% of data scientist postings. Cloud and infrastructure skills command a premium. The gap between a generalist analyst and a specialist engineer is wider here than in almost any field.

What's genuinely hard about it

Entry competition is fierce, and employers consistently prefer two to four years' experience — which is the classic problem of needing experience to get experience.

The skill bar keeps rising. What qualified someone as a data scientist in 2020 now qualifies them as an analyst. Programming expectations have moved down the ladder from ML engineers to analysts, and that migration is continuing.

The title inflation is genuinely confusing. "Data analyst," "data scientist," "analytics engineer" and "ML engineer" describe overlapping work with enormous pay differences, and it is hard for a newcomer to judge which is which from a job posting.

And there is no structural floor. No license, no physical requirement. Whatever protection you have, you earned personally.

Who this work suits — and who it doesn't

Data science tends to suit people who:

- **Are genuinely curious about the question, not the technique.** The best predictor of who moves up rather than out.
- **Can hold statistical rigour and commercial context simultaneously**
- **Are comfortable saying an answer is uncertain** — much of the value is knowing what the data can't tell you
- **Keep learning tooling without resenting it**, because it changes constantly
- **Can communicate to people who don't want the detail**

It's harder for people who:

- Want **defined, repeatable tasks** — that's the automating layer
- Are drawn to the **technical craft** more than the problem
- Need **stable tooling** across a career
- Chose it because it seemed like the safe technical option. That reasoning is precisely what the section on proximity to AI is about.

If some of that second list sounds like you, it isn't a reason to abandon the field — it's a reason to steer toward the parts of it that fit, and toward the building end, which is what the rest of this does.

What else is moving this market

AI adoption is itself the demand driver. 88% of organisations now use AI in at least one function, and every one of them needs people who can build and maintain the data infrastructure underneath it. That's a genuine tailwind, and it's the same shape as electrical engineering's — the buildout creating demand for the people who enable it.

Geography has shifted. New York has overtaken California as the leading location for data science postings, and 40–50% of roles offer remote options.

And domain specialization is where the premium sits — healthcare, finance and energy all pay above general technology for the same technical skill.

The AI-native advantage — what it actually looks like

The situation: this is the field where AI tooling arrived first and most completely, which means the bar for what counts as valuable has already moved. Being already in it, close to the work, is a real advantage here — you can build this fluency inside a live internship, a research project or a first job, which is exactly where it counts.

WHAT AN AI-NATIVE DATA PROFESSIONAL LOOKS LIKE

- 1. Frame the question.** Analysis is now cheap; knowing what to analyze is not. This is the single most protective skill in the field.
- 2. Verify.** Knowing when a generated analysis is confidently wrong — which requires knowing the business and the data well enough to smell an implausible result.
- 3. Build rather than run.** ML engineering and data infrastructure score better than analysis because building systems is a different kind of work from using them.

4. Own a recommendation. Everything in this guide points at accountability. Anything that puts you closer to a decision you answer for moves you up the scale.

5. Know what shouldn't be automated — where a model will encode a bias, or where a statistical answer is the wrong kind of answer.

All five are available to you now — not after some future qualification. Turning them into concrete moves is the rest of this guide.

Where a data science degree can take you later

Broad and well-regarded: data science and ML engineering, data and analytics engineering, research science, quantitative finance, product analytics, AI/ML infrastructure, technical leadership, and domain-specialist roles across healthcare, energy and government.

The pattern in this index holds: roles where someone builds a system or owns a recommendation score better than roles that produce analysis for someone else to act on. That pattern is the compass for every choice below.

Repositioning from here

The program-selection stage is behind you. The useful questions now are different — and the good news bears repeating: almost none of your exposure is fixed yet. Data science is a 2.2-point spread from 6.0 to 8.2, and where you land inside it is decided mostly by things still in front of you. Nearly all of them point the same way: toward building and ownership, and away from generalist querying.

If you're still in the degree or training

Three levers, in order of leverage.

1. Aim at the building end, not the querying end. Generalist entry analysis (8.2) and analytics-engineer / BI-developer work (7.6) are the exposed rungs — the report-and-dashboard layer, first in line for automation. The protected destinations — ML and AI engineering (6.0), senior data science (6.5), data engineering (7.0) — are reached by learning to build systems, not just use them. If you haven't concentrated your electives and projects there, this is the single highest-leverage decision left in the degree.

2. Statistical depth, while it's still free. Statistical judgment survives the automation of running the analysis; tooling fluency does not. Knowing why a method applies outlasts knowing which button to press — and every serious statistics, experimental-design and inference module you can still take is a step from the exposed end toward the protected one, far cheaper to acquire now than to bolt on later.

3. A domain, and real messy data. Pair the work with healthcare, finance, energy or biology: the domain is what lets you know a result is wrong, and it's what those sectors pay a premium for. That domain knowledge, applied to real messy data through an internship or research project, is the experience that separates the protected end from the exposed floor — and in a field that prefers demonstrated experience over credentials more than almost any other, it's the closest thing to a license.

If you've graduated and you're job-hunting

Here the title on the offer matters more than it looks, because the titles overlap wildly and the exposure underneath them does not.

- **"Data analyst" is the 8.2**, and "analytics engineer / BI developer" the 7.6 — the default graduate destinations and the automating layer at once. Taking one isn't fatal — but take it as a deliberate stepping-stone with an exit toward building or ownership in mind, not as the plan.
- **Aim at the building tracks (6.0–7.0).** ML and AI engineering, data engineering, research and senior data science — more protected, because building a system is different work from producing analysis someone else acts on.
- **Attach to a domain, not "data in general."** Healthcare, finance and energy pay above general technology for the same technical skill, and the domain is what lets you catch a wrong result. "Open to any data role" aims straight at the exposed middle.
- **Chase the seat that builds or owns.** Anything where you build a system or own a recommendation that costs money if it's wrong — even at a smaller organisation, where that responsibility arrives sooner — moves you up the scale.

The AI-native move, from where you stand

You can't undo the tasks that are already automating, but you can be the entrant who directs these tools rather than competes with them. The five capabilities above — frame the question, verify, build rather than run, own a recommendation, know what shouldn't be automated — are all available right now, inside a live project, an internship or a first job. The most powerful for someone already in it is the first: be the person who decides what question is worth asking, because analysis is now cheap and framing is not. That judgment requires knowing the business

and the data well enough to smell an implausible result — and you can start building it this term, on real messy data, not after some future qualification.

The career-hunting checklist

The analysis tells you where the exposure is. This is what to do about it — the part that's actually in your hands. Most of the protection in data science is built here, in these habits and choices, rather than handed to you. The first list is for anyone still in the degree; the second is for the job hunt itself.

During the degree — build what can't be automated

- **Aim at the building end, not the querying end.** ML and AI engineering (6.0), senior data science (6.5) and data engineering (7.0) are the protected tracks; analytics-engineer / BI-developer work (7.6) and generalist entry analysis (8.2) are the exposed ones. Point every elective and project at the former.
- **Take statistics and mathematics seriously — they're the durable layer.** Statistical judgment survives the automation of running the analysis; tooling fluency does not. Knowing why a method applies outlasts knowing which button to press.
- **Get real experience on messy data before anyone asks for it.** An internship, a research project, a competition — anything that isn't a clean teaching set. In a field that prefers demonstrated experience over credentials more than almost any other, this is the closest thing to a license.
- **Pair the work with a domain.** Healthcare, finance, energy, biology. The domain is what lets you know a result is wrong, and it's what those sectors pay a premium for — machine learning already appears in 77% of data scientist postings, so raw tooling alone won't differentiate you.
- **Learn to build systems, not just use them.** ML engineering and data infrastructure score better precisely because building is different work from running. A program that only teaches Python, SQL and dashboarding is training you for the 8.2 rung.
- **Feed curiosity and adaptability on purpose.** The language, the library and the platform will all turn over repeatedly; the person who re-tools fastest, and stays curious about the question rather than the technique, compounds.
- **Keep the frame wide.** Vet statistics or mathematics as the degree with data on top, computer science with an ML specialization, and domain pairings before committing.

The most crowded lane is the tooling-focused data science degree, and it points at the automating work.

During the job hunt — aim, don't just apply

- **Target the protected lanes on purpose.** ML and AI engineering, data engineering, senior and research data science, domain-specialist roles — not "data analyst, open to anything," which points straight at the 8.2.
 - **Read the title, not just the offer.** "Data analyst," "data scientist," "analytics engineer" and "ML engineer" describe overlapping work with enormous pay differences and very different exposure. If you take a generalist analyst role, take it as a deliberate stepping-stone with an exit toward building or ownership in mind, not as the plan.
 - **Don't grab a role just because it's on offer in an exposed lane.** A role being available doesn't make it safe. An employer still running a large generalist-analyst intake may simply be behind the curve — and that's exactly the rung where the skill bar is rising fastest underneath the title. Availability can be a lagging indicator, not a green light.
 - **Prefer the seat that builds or owns.** A role where you build a system or own a recommendation someone acts on — even at a smaller organisation, where that responsibility arrives sooner — puts you closer to the protected end than one producing analysis for someone else to act on.
 - **Ask the employer where AI is heading in the role.** Whether they're building the job around framing, verification and system-building, or waiting to be forced to, tells you if the seat grows or is quietly being hollowed out.
 - **Weigh trajectory over the starting salary.** Entry analyst pay is genuinely high right now — reported around \$90,000 and rising — but a rising salary on an exposed rung is not the same as a protected one. Choose the lane that ages best, not the highest first-year number.
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Go deeper — further reading

- **Data Scientists — Occupational Outlook Handbook** — US Bureau of Labor Statistics. The primary source for the growth projection, median pay and annual openings. Free, and the single most important counterweight to our own score. Read it before accepting anything in this guide.
- **Data scientist job outlook 2026** — based on analysis of over 1,000 job postings. Useful for what employers actually ask for — including that machine learning now appears in 77%

of data scientist postings, and how quickly skill requirements are migrating down the ladder.

- **Data analyst job outlook 2026** — the same analysis for the analyst tier, including the rise in entry-level salaries and the shift toward specialists over generalists.

The counterargument — it's near the front, and it's the strongest in this index

We've placed it near the front rather than here, because a reader deserves to weigh it before reading four thousand words built on the opposite view. In summary: BLS projects 33.5% growth, the WEF ranks this the fastest-growing job in percentage terms through 2030, McKinsey names data roles the most in-demand AI hires, and entry-level salaries are rising.

If you conclude from that evidence that our 8.2 is wrong, that is a defensible reading and we would rather you reached it with the data in front of you than take our number on trust.

Bottom line

Data science is the guide where our framework and the labor market disagree most, and we've tried to give you both — which is exactly why being already in it is not a problem: the decisive moves were always going to happen from here.

Our analysis says the entry tier is among the most exposed work we measure — 8.2, with a job description that reads like a list of automated tasks, and no license or physical requirement anywhere in the field to slow it down.

The labor market says this is the fourth-fastest-growing occupation in the economy, with salaries rising rather than falling.

What we think reconciles them: the title is surviving while its contents change completely. Demand for people who can work with data is growing fast. Demand for people who can only query, clean and chart is not. The skill bar has risen sharply and keeps rising — which is why standing still on the generalist rung is the one thing not to do.

So the useful advice isn't a verdict on the field — it's a direction. Take statistics seriously rather than tooling, because statistical judgment survives the automation of running the analysis. Steer toward engineering and ML rather than analysis. Pair it with a domain — that's what lets you know when an answer is wrong, and it's where the premium sits. And get real

experience with messy data now, because employers here prefer experience over credentials more than in almost any field.

And one thing worth saying plainly, if you're in this because it seemed like the safe technical option: being close to AI does not protect you from it. Exposure follows the shape of the work, not its subject. The protected end of this field is real, and it's still available to you from where you stand — it's just further up than the brochures suggested.

Questions worth sitting with

Whether you're the one in it or a parent thinking it through, these are the conversations that move the needle now. Read "you" as whoever the decision is about.

Where you actually stand

1. Which track are you on — or aiming at — and does it point at *building and ownership* (ML engineering, data engineering, owning a recommendation) or at *querying* (generalist analysis, dashboarding)? 2. Looking at the score table, which destination are you currently pointed at, and is it the one you'd choose on purpose? 3. Is there anything you've built or owned end-to-end — a system, a model in production, a recommendation someone acted on — or only analyses you handed off? If it's the latter, what's the nearest way to change that?

The move from here

4. What's the single most protected role you could realistically aim for from where you are now, and what would make you a credible candidate for it? 5. If you're still studying: what's the most rigorous statistics or systems-building module still open to you, and is there a reason not to take it? 6. If you're job-hunting: are you aiming at a building track on purpose, or applying to whatever's open? Name the three roles you're actually chasing, and whether they build or only query.

The AI question, practically

7. Of the five capabilities — frame the question, verify, build rather than run, own a recommendation, know what shouldn't be automated — which one could you visibly demonstrate within the next few months? 8. Can you point to a moment where you caught a generated analysis that was confidently wrong? If not, where's the easiest place to start building that instinct — and the domain knowledge it rests on?

Testing it against reality

9. **The most valuable single action:** work with real, messy data on a real question. A project, a competition, anything that isn't a clean teaching set. 10. If the first offer that comes is a generalist "data analyst" role, what's your plan — take it as a deliberate stepping-stone with an exit toward building in mind, or hold out for something closer to engineering? Decide before the offer, not after.

This is analysis and informed judgment about a fast-moving situation. For data science specifically, our score faces the strongest counter-evidence of any career in this index, and we have set that evidence out near the front rather than at the end. A reader who weighs it differently from us is reaching a defensible conclusion.

Scores for 2023 and 2025 are retrospectively calculated using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed, and not how many people employers want to hire.

Technical scoring appendix — Data Science

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

1. The formula

$$\begin{aligned}\text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment})\end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-eight careers, which is what makes the scores comparable.

Task automatability (A) — *what share of the working week could a capable current system already do to an acceptable standard?*

RATING	ANCHOR
0–2	Almost nothing. The work is physical, relational or wholly novel.
3–4	Administrative surround only — notes, scheduling, correspondence.
5–6	A meaningful minority of the week, mostly documentation and lookup.
7–8	A large share, including tasks central to the junior version of the role.
9–10	Most of the described duties, at or near professional standard.

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
0–2	Protected by statute or physical necessity. Training hours cannot be automated.
3–4	Stable. Some junior tasks absorbed; the apprenticeship survives.
5–6	Visible narrowing. Employers prefer experience; junior intake reduced.
7–8	Substantial. The work juniors learned on has largely gone.
9–10	The training layer and the automatable layer were the same layer.

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
0–2	Fully screen-based. Location-independent.
3–4	Occasional presence; the core work is not physical.
5–6	Regular physical presence, in largely controlled conditions.
7–8	Substantially physical, with some variability of setting.
9–10	Irreducibly physical, in conditions that differ every time.

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human?*

RATING	ANCHOR
0–2	No license exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent, where being wrong is costly?*

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

ML / AI ENGINEER — 6.0 (MODERATE-HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.3	5.4	6.5	2.27
Entry-path erosion	15%	3.5	4.2	5.0	0.75
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	6.5	6.5	6.5	0.53
Regulatory / licensing	10%	1.0	1.0	1.0	0.9
Judgment under novelty	10%	8.0	8.0	8.0	0.2
				Total	6.0 → 6.0

SENIOR DATA SCIENTIST — 6.5 (MODERATE-HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.0	6.4	7.2	2.52
Entry-path erosion	15%	4.0	4.8	5.5	0.82
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	6.0	6.0	6.0	0.6
Regulatory / licensing	10%	1.0	1.0	1.0	0.9
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	6.49 → 6.5

DATA ENGINEER — 7.0 (HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.7	6.8	7.9	2.77
Entry-path erosion	15%	4.5	5.2	6.0	0.9
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	5.5	5.5	5.5	0.67
Regulatory / licensing	10%	1.0	1.0	1.0	0.9
Judgment under novelty	10%	6.0	6.0	6.0	0.4
				Total	6.99 → 7.0

ANALYTICS ENGINEER / BI DEVELOPER — 7.6 (HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.7	7.3	8.4	2.94
Entry-path erosion	15%	5.5	6.5	7.2	1.08
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	4.5	4.5	4.5	0.82
Regulatory / licensing	10%	1.0	1.0	1.0	0.9
Judgment under novelty	10%	5.0	5.0	5.0	0.5
				Total	7.59 → 7.6

ENTRY DATA ANALYST — 8.2 (HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.9	7.7	9.4	3.29
Entry-path erosion	15%	6.0	7.0	7.8	1.17
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	4.0	4.0	4.0	0.9
Regulatory / licensing	10%	1.0	1.0	1.0	0.9
Judgment under novelty	10%	4.0	4.0	4.0	0.6
				Total	8.21 → 8.2

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

ML / AI engineer

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	4.3	3.5	2.98	5.0
2025	5.4	4.2	2.98	5.5
Fall 2026	6.5	5.0	2.98	6.0

Movement decomposition: automatability contributed **+0.77** and entry-path erosion **+0.22**, for a total of **+1.0** points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labor market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.
- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.